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老年COPD患者的谵妄 机械通气合并呼吸衰竭:一项前瞻性 队列研究

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摘要

背景:COPD合并呼吸衰竭在重症监护病房(ICU)非常常见。目的探讨老年COPD机械通气患者谵妄的现状及其影响因素。

方法:选择我院ICU收治的COPD合并呼吸衰竭机械通气患者。对入选患者的特征进行评估。采用Pearson相关分析评价患者与谵妄的特征。通过Logistic回归分析老年COPD合并呼吸衰竭机械通气患者谵妄的危险因素。

结果:共纳入237例COPD合并呼吸衰竭患者, 谵妄发生率为21.94%。Pearson相关分析显示, 年龄($r = 0.784$)、BMI ($r = 0.709$)、高血压($r = 0.696$)、APACHE II评分($r = 0.801$)、CPOT ($r = 0.513$)、镇静($r = 0.626$)、 $-PaO_2$ ($r = 0.611$)与谵妄的发生均有相关性($p < 0.05$)。Logistic回归分析显示:年龄 ≥ 75 岁(OR 3.112, 95% CI 2.144-4.602)、BMI指数 ≤ 19 kg/m² (OR 2.742, 95% CI 1.801-3.355)、高血压(OR 1.909, 95% CI 1.415-2.421)、APACHE II评分 ≥ 15 (OR 2.087, 95% CI 1.724-2.615)、CPOT评分 ≥ 5 (OR 1.778, 95% CI 1.206-2.641)、镇静(OR 3.147, 95% CI 2.714-3.758)、 $-PaO_2 \leq 75$ mmHg(OR 2.196, 95% CI 1.875 ~ 3.088)是老年COPD合并呼吸衰竭机械通气患者发生谵妄的危险因素(均 $p < 0.05$)。结论:谵妄在COPD合并呼吸衰竭机械通气患者中较为常见, 且影响因素较多。医护人员应更加重视患者的危险因素, 尽早采取干预措施, 降低谵妄的发生率。

关键词:COPD, 呼吸衰竭, 机械通气, 老年人, 谵妄, 护理, 管理, 预防

背景

呼吸衰竭的发病率逐年上升, 其中慢性阻塞性肺疾病(COPD)引起的呼吸衰竭尤为显著。COPD患者往往具有肺基础功能差、病程长、易复发、多系统病变[1]等特点。

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呼吸衰竭是人们住进重症监护病房(ICU)的常见原因。既往研究[3-5]表明,机械通气是治疗呼吸衰竭的有效手段,对COPD合并呼吸衰竭患者的治疗效果显著。机械通气患者可产生一系列心理反应,导致ICU内认知功能改变或知觉障碍、谵妄[6,7]。既往研究结果[8-10]表明,机械通气患者谵妄的发生率可高达60.48%。一旦发生谵妄,患者可能出现住院时间延长、机械通气脱机困难、认知功能障碍、死亡率增高等[11,12]。

谵妄是老年患者常见但又容易被忽视的一种异常精神状态。其表现为一系列症状,其特征为不同程度的认知障碍、缺乏方向感、思维混乱,形成[13]综合征。谵妄常以明显的认知功能障碍和意识水平改变为基本临床特征。部分患者还可能伴有定向、注意、知觉、睡眠周期等方面的异常[15]。谵妄的预防和早期护理对机械通气呼吸衰竭患者的预后至关重要[16,17]。目前,老年COPD合并呼吸衰竭机械通气患者谵妄的危险因素尚不清楚。因此,我们旨在评价老年COPD合并呼吸衰竭患者机械通气的特点,明确机械通气呼吸衰竭患者谵妄的危险因素,为COPD的临床管理和护理提供参考。

方法

伦理

在本研究中,所有方法均按照相关指南和规定进行。本研究采用前瞻性队列设计。本研究方案已经我院伦理委员会审核批准(批准号:IC19060048)。此外,已获得纳入患者所有亲属的知情同意。

患者

本研究选择我院ICU收治的COPD急性加重期继发机械通气患者作为研究人群。患者入组标准为:①患者年龄 ≥ 65 岁;②诊断为C

OPD并发呼吸衰竭的患者;③机械通气超过24 h的患者,患者的排除标准为:①入院时有精神病史或意识受损且格拉斯哥昏迷量表(GCS)评分 ≤ 6 分的患者;②采用无创通气(NIV)治疗的患者也排除在外;③由于机械通气持续时间,住院期间实际死亡的患者可明显受到死亡率的偏倚;本人或家属不愿参与研究的患者。

样本量计算

本研究的样本量根据两组比较公式计算[18,19]:

$n = 2 \arcsin \sqrt{\frac{1}{2}}$, 我们假

设 $P_{\max} - \arcsin P_{\min}$

$\left(\frac{1}{2} \right)$

$\alpha = 0.05$, $\beta = 0.1$, $v = 3 - 1 = 2$, $\lambda = 8.86$, 设 $P_{\max}$, $P_{\min}$ 分别为COPD患者谵妄发生率的最大值和最小值(38.26%和6.02%)[20], 则得到 $n \approx 182$ 的结果。同时,考虑到10%左右患者的潜在损失率,共纳入201例患者。我们尝试尽可能多的纳入患者。

谵妄的诊断

在本研究中,在重症监护病房[21]的混乱评估方法被用于评估谵妄患者在整个ICU住院期间。CAM-ICU包括4个方面:①精神状态突然变化或波动,②注意力不集中,③意识水平变化,④思维紊乱。当患者同时满足①、②、③或④时,可诊断为谵妄。我们分别于每天8:00、16:00、22:00对患者进行谵妄评估。

数据收集

在研究开始前,我们对研究者进行了统一的培训,并向患者及其家属说明了研究的细节,以获得合作。我们通过询问患者家属、查阅病历、使用量表等方式收集患者的临床资料。我们收集的数据包括年龄、性别、BMI、高血压、糖尿病、高脂血症、贫血、急性生理和慢性健康评分(APACHE II)、临界疼痛观察工具(CPOT)[22]、镇静使用情况、机械通气持续时间。CPOT是为无法沟通的ICU患者开发的。每日8:00采用CPOT量表评估患者疼痛状态,包括面部表情、体力活动、肌肉张力、机械通气依从性4个项目。每个项目打分为0~2分,且

总分从0分到8分不等。分数越高，说明患者疼痛越严重。

此外，我们收集了机械通气前的实验室检查结果，包括:白蛋白、血红蛋白、血糖、肌酐、尿素氮、三碘甲状腺原氨酸(T3)、甲状腺素(T4)、促甲状腺激素(TSH)、游离三碘甲状腺素- 9 (FT3)、游离甲状腺素(FT4)、动脉氧分压- (PaO₂)、动脉二氧化碳分压(PaCO₂)水平。

统计学方法

本研究采用SPSS 22.0统计软件进行数据分析。枚举数据以百分数表示，组间比较采用卡方检验。计量资料以均数±标准差表示，组间比较采用t检验。采用Pearson相关分析评价患者与谵妄的特征。通过Logistic回归分析老年COPD合并呼吸衰竭患者机械通气后谵妄的危险因素。组间比较，P < 0.05，差异有统计学意义。

结果

纳入COPD患者的特征

共纳入237例COPD合并呼吸衰竭患者，其中52例出现谵妄，谵妄发生率为21.94%。如表1所示，谵妄组与非谵妄组在年龄、BMI、高血压、APACHE II评分、CPOT、镇静、机械通气时间等方面差异均有统计学意义(p < 0.05)。谵妄组与非谵妄组在性别、

糖尿病、高脂血症、贫血等方面差异无统计学意义(均p < 0.05)。

机械通气前实验室检查结果。

如表2所示，谵妄组与无谵妄组p - ao₂差异有统计学意义(p = 0.016)。两组患者白蛋白、血红蛋白、血糖、肌酐、尿素氮、pH、T3、T4、TSH、FT3、FT4、- paco₂差异均无统计学意义(p < 0.05)。

Pearson相关分析

如表3所示，Pearson相关分析显示，年龄(r = 0.784)、BMI(r = 0.709)、 高 血 压 (r = 0.696)、APACHE II评分(r = 0.801)、CPOT(r = 0.513)、镇静(r = 0.626)、-PaO₂ (r = 0.611)与谵妄的发生相关(均p < 0.05)。

Logistic回归分析

多变量logistic回归的变量赋值情况见表4。如表5所示，Logistic回归分析显示:年龄≥75岁(OR 3.112, 95% CI 2.144-4.602)、BMI 指 数 ≤19 kg/m² (OR 2.742, 95% CI 1.801-3.355)、高血压(OR 1.909, 95% CI 1.415-2.421)、APACHE II评分≥15(OR 2.087, 95% CI 1.724-2.615)、CPOT 评 分 ≥5(OR 1.778, 95% CI 1.206-2.641)、镇静(OR 3.147, 95% CI 2.714-3.758)、-PaO₂ ≤75 mmHg(OR 2.196、95% CI 1.875 ~ 3.088)是老年COPD合并呼吸衰竭机械通气患者发生谵妄的危险因素(p < 0.05)。

表1纳入COPD患者的特征

Variables	Delirium group(n = 52)	No delirium group(n = 185)	t/χ ²	p
Age(y)	78.11 ± 10.26	69.84 ± 9.03	9.445	0.016
Male/female	31/21	110/75	2.061	0.093
BMI (kg/m ²)	18.12 ± 4.67	20.05 ± 4.19	3.142	0.007
Hypertension	40(76.92%)	66(35.68%)	1.065	0.024
Diabetes mellitus	25(48.08%)	84(45.41%)	3.129	0.108
Hyperlipidemia	20(38.46%)	67(36.22%)	2.025	0.082
Anemia	28(53.85%)	92(49.73%)	1.113	0.104
APACHE II score	17.49 ± 4.07	12.62 ± 3.28	5.108	0.008
CPOT	6.14 ± 2.01	4.21 ± 1.66	2.045	0.015
Sedation	41(78.85%)	59(31.89%)	1.251	0.002
Duration of mechanical Ventilation(h)	73.14 ± 22.17	68.05 ± 23.14	11.038	0.048

身体质量指数，APACHE II急性生理和慢性健康评分，CPOT临界疼痛观察工具

表2机械通气前实验室检查结果

Variables	Delirium group(n = 52)	No delirium group(n = 185)	t/ χ^2	p
Albumin(g/L)	41.91 ± 8.02	42.88 ± 10.24	6.132	0.081
Hemoglobin(g/L)	123.63 ± 17.45	125.12 ± 13.44	13.562	0.103
Blood sugar(mmol/L)	7.46 ± 1.27	7.51 ± 1.35	1.241	0.196
Creatinine(μ mol/L)	76.53 ± 9.82	74.17 ± 8.95	5.033	0.094
Urea Nitrogen(mmol/L)	7.72 ± 1.25	7.49 ± 1.47	2.204	0.176
pH	7.38 ± 1.15	7.40 ± 1.02	3.847	0.113
T3(mU/L)	1.42 ± 0.35	1.36 ± 0.28	1.012	0.095
T4(mU/L)	96.05 ± 12.15	97.24 ± 14.02	4.225	0.061
TSH(mU/L)	2.42 ± 0.35	2.63 ± 0.53	1.607	0.084
FT3(mU/L)	3.18 ± 0.86	3.26 ± 0.95	1.874	0.262
FT4(mU/L)	16.94 ± 5.64	16.66 ± 5.13	2.084	0.114
PaO ₂ (mmHg)	71.21 ± 19.05	79.41 ± 15.89	10.123	0.016
PaCO ₂ (mmHg)	34.01 ± 7.24	32.54 ± 8.38	5.019	0.052

T3三碘甲状腺原氨酸、T4甲状腺素,TSH促甲状腺激素, FT3游离三碘甲状腺原氨酸, FT4游离甲状腺素, PaO₂动脉血氧分压, PaCO₂动脉血二氧化碳分压

表4多变量logistic回归的变量赋值情况

表3患者特征与谵妄的Pearson相关分析

Variables	r	P
Age(y)	0.784	0.006
Gender	0.016	0.112
BMI (kg/m ²)	0.709	0.014
Hypertension	0.696	0.045
Diabetes mellitus	0.123	0.079
Hyperlipidemia	0.126	0.205
Anemia	0.165	0.148
APACHE II score	0.801	0.025
CPOT	0.513	0.036
Sedation	0.626	0.029
Duration of mechanical ventilation(h)	0.265	0.103
Albumin(g/L)	0.163	0.088
Hemoglobin(g/L)	0.124	0.105
Blood sugar(mmol/L)	0.168	0.059
Creatinine(μ mol/L)	0.042	0.146
Urea nitrogen(mmol/L)	0.146	0.093
T3(mU/L)	0.113	0.119
T4(mU/L)	0.147	0.101
TSH(mU/L)	0.073	0.162
FT3(mU/L)	0.187	0.067
FT4(mU/L)	0.153	0.122
PaO ₂ (mmHg)	0.611	0.021
PaCO ₂ (mmHg)	0.195	0.089

BMI身体质量指数、APACHE II急性生理和慢性健康评分、CPOT危重疼痛观察工具、PaO₂动脉氧分压、T3三碘甲状腺原氨酸、T4甲状腺素,TSH促甲状腺激素, FT3游离三碘甲状腺原氨酸, FT4游离甲状腺素, PaO₂动脉血氧分压, PaCO₂动脉血二氧化碳分压

Factors	Variables	Assignment
Delirium	Y	Yes = 1, no = 2
Age(y)	X ₁	≥ 75y = 1, < 75y = 2
BMI(kg/m ²)	X ₂	≤ 19 = 1, > 19 = 2
Hypertension	X ₃	Yes = 1, no = 2
APACHE II score	X ₄	≥ 15 = 1, < 15 = 2
CPOT	X ₅	≥ 5 = 1, < 5 = 2
Sedation	X ₆	Yes = 1, no = 2
PaO ₂ (mmHg)	X ₇	≤ 75 = 1, > 75 = 2

身体质量指数、APACHE II急性生理和慢性健康评分、CPOT临界疼痛观察工具、PaO₂动脉氧分压

表5老年COPD合并呼吸衰竭机械通气患者谵妄危险因素 Logistic回归分析

Items	β	SE	OR	95%CI	p
Age ≥ 75y	0.113	0.201	3.112	2.144–4.602	0.016
BMI ≤ 19 kg/m ²	0.215	0.316	2.742	1.801–3.355	0.024
Hypertension	0.164	0.293	1.909	1.415–2.421	0.032
APACHE II score ≥ 15	0.286	0.541	2.087	1.724–2.615	0.007
CPOT ≥ 5	0.195	0.426	1.778	1.206–2.641	0.023
Sedation	0.167	0.541	3.147	2.714–3.758	0.017
PaO ₂ ≤ 75 mmHg	0.118	0.377	2.196	1.875–3.088	0.009

BMI体重指数, APACHE II急性生理和慢性健康评分, CPOT临界疼痛观察工具, PaO₂动脉氧分压

讨论

呼吸衰竭患者因病情严重、发展迅速，常出现多种并发症，谵妄的风险特别高[23,24]。近年来，对ICU谵妄的研究逐渐深入和完善。ICU谵妄发生率为10.13-78.46%，其中机械通气患者居多[25,26]。目前关于COPD合并呼吸衰竭患者机械通气[27]谵妄现状的研究尚缺乏。本研究结果表明，COPD合并呼吸衰竭机械通气患者谵妄发生率为21.94%，年龄 ≥ 75 岁、BMI ≤ 19 kg/m²、高血压、APACHE II评分 ≥ 15 、CPOT ≥ 5 、镇静、P-aO₂ ≤ 75 mmHg是老年COPD合并呼吸衰竭机械通气患者发生谵妄的危险因素，需要对患者进行早期有针对性的干预和护理，以减少谵妄的发生。

高龄是谵妄发生的独立危险因素。COPD患者多为老年患者。谵妄的原因是神经递质功能异常，血脑屏障受损[28-30]。老年人大脑对外部刺激的接受性降低，炎症反应增强，表现为随着年龄的增长，老年人大脑细胞进行性凋亡，脑功能下降，脑内神经递质合酶活性降低[31,32]。此外，老年患者脑内小胶质细胞活化更为明显，神经元微环境发生改变，炎症因子水平升高。在机械通气的刺激下，可产生更强、持续时间更长的炎症反应，可引起脑内神经递质异常，损伤血脑屏障，导致谵妄的发生[33,34]。

本研究结果发现，BMI低于19 kg/m²是老年COPD合并呼吸衰竭患者谵妄的重要因素。已有多项研究[35-37]表明BMI < 20 kg/m²是术后谵妄的危险因素，这与我们的研究结果一致。较低的BMI与不良的营养状况有关。对于营养状况较差的患者，择期手术前建议进行营养支持治疗，积极纠正低蛋白血症，调整营养状态，可能有助于减少术后谵妄的发生[39,40]。

APACHE II评分代表疾病的严重程度，评分与疾病的严重

程度呈正相关。患者APACHE II评分越高，疾病引起的应激反应越严重，高级神经中枢活动急性紊乱越明显，越容易出现谵妄[41]。因此，医护人员在重视疾病治疗的同时，应综合评估患者病情，密切观察患者意识、血氧饱和度、疾病表现、实验室指标等，及时采取措施[42]。此外，预测性评估对于有效预测此类患者的潜在风险，并立即采取措施，避免病情恶化[43]尤为重要。

COPD合并呼吸衰竭的患者通常有-pao₂下降。-pao₂越低，患者缺氧越严重，内环境紊乱越明显。COPD患者脑组织供氧减少。P-aO₂越低，供氧失衡越明显，脑组织水肿[45]越严重。此外，-PaO₂越低，气短、喘鸣等症状越明显，全身不适越强烈，常伴有大量出汗、血压下降、心率加快等，患者易产生焦虑、恐惧、易怒等负面心理。甚至出现精神行为，进而诱发谵妄[46,47]。有研究[48,49]指出，血氧饱和度降低和低氧血症的患者更容易出现谵妄。因此，医护人员应更加重视-pao₂ < 75 mmHg的患者，尤其要警惕组织灌注不足的患者。在条件允许的情况下，医务人员应尽快拔除气管插管，拔管后，根据患者病情使用无创通气或鼻插管吸氧，不同类型的呼吸衰竭给予不同的供氧方式[50]。此外，COPD患者通常通过增加呼吸次数来补偿。这种补偿主要是胸呼吸[51]。胸式呼吸的效果要低于腹式呼吸[52]。患者会感到不同程度的疲劳。因此，需要鼓励患者在拔管后进行呼吸练习。由于气管插管、克制、与亲人分离、焦虑恐惧等各种因素，各种程度的疼痛应该由医疗人员来加重。疼痛使患者的应激反应加重，发生一系列的生理和心理变化[53,54]。医护人员应重视疼痛对谵妄的影响，进行日常疼痛评估，观察患者是否有生理上的

以及心理变化,并尽快实施干预措施,减轻患者的疼痛,防止谵妄[55]的发生。对于有疼痛的患者,评估疼痛的部位、伴随症状、持续时间,按照医生的指示进行镇痛,并观察用药效果[56,57]。此外,医务人员应根据效果调整用药剂量,避免剂量过少或过量。镇痛药剂量过少会达不到止痛效果,过量会引起呼吸抑制[58]等不良反应。因此,加强对这类患者疼痛及镇痛效果的评价就显得尤为重要。

镇静是COPD和呼吸衰竭患者接受机械通气的常用治疗方法。镇静可引起颅脑神经系统功能障碍,影响对刺激的反应,诱发谵妄[59]。ICU常用的临床镇静剂包括异丙酚、盐酸右美托咪定和咪达唑仑。其中丙泊酚可影响脑组织灌注,易引起低血压和呼吸抑制[60]。咪达唑仑常见的不良反应有嗜睡、头痛、共济失调、幻觉等,可延长ICU住院时间和机械通气时间,增加谵妄的发生率[61]。如果突然停药,会引起戒断症状,因此应逐渐减少剂量,并使用苯二氮卓类拮抗剂氟马西尼逆转过量[62]。在ICU中,右美托咪定比其他镇静剂更不容易引起谵妄[63-65]。盐酸右美托咪定镇静作用轻微,适用于机械通气患者。患者易苏醒,但易引起低血压[66]。此外,盐酸右美托咪定增加迷走神经的刺激,可引起心动过缓[67]。既往研究[68,69]指出,盐酸右美托咪定对机械通气患者具有良好的镇静作用,可减少谵妄的发生,缩短谵妄持续时间。镇静患者应实行每日镇静中断,避免镇静药物过度使用,减少药物蓄积,便于患者早期拔管[70]。

必须考虑到本研究的几个局限性。首先,镇静是谵妄发展的重要因素,在我们的研究中镇静并不常见,并且本研究中用于镇静的药物不同,谵妄与镇静药物使用的关联需要在未来更大的样本量中进行更多的研究。其次,我们没有

入院时COPD严重程度的数据,这可能与谵妄的发生有关。今后有必要进一步评价COPD严重程度与谵妄的关系。第三,这是一项单中心研究,因此其泛化性可能受到限制。即使已经计算了样本量的统计能力来检测组间差异,未来还需要更多的多中心、更大样本量的研究来进一步阐明COPD患者谵妄的影响因素。

结论

综上所述,本研究发现,COPD合并呼吸衰竭机械通气患者谵妄发生率为21.94%,且年龄 ≥ 75 岁、BMI ≤ 19 kg/m²、高血压、APACHE II评分 ≥ 15 、CPOT ≥ 5 、镇静、P为谵妄危险因素。因此,医护人员应多关注有危险因素的患者,尽早采取干预措施,减少谵妄的发生,避免谵妄相关不良事件的发生。但受样本量限制,需要更多综合变量的研究来进一步评价谵妄的影响因素,为COPD患者谵妄的临床管理和预防提供参考。

缩写词

COPD:慢性阻塞性肺疾病;ICU:重症监护病房;CAM-ICU:ICU混淆评定量表;BMI:身体质量指数;GCS:格拉斯哥昏迷量表;APACHE II期:急性生理和慢性健康评分;CPOT:临界疼痛观察工具;T3:三碘甲状腺原氨酸;T4:甲状腺素;TSH:促甲状腺激素;FT3:游离三碘甲状腺原氨酸;FT4:游离甲状腺素;PaO₂:动脉血氧分压;PaCO₂:动脉二氧化碳分压。

致谢

无。

作者贡献

FM设计研究;XF、LW、GW、XL、XW、SM、FM进行了研究;XF、LW分析数据;XF、LW、FM撰写稿件初稿;fm对最终的内容负有主要责任。所有作者阅读并审定终稿。

经费

无。

数据和材料的可获得性

本研究过程中产生或分析的所有数据都包含在这篇发表的文章中。

声明

伦理批准并同意参与

在本研究中,所有方法均按照相关指南和规定进行。本研究采用前瞻性队列设计。研究方案已经我院伦理委员会审核批准(批准号:IC19060048)。此外,已获得纳入患者所有亲属的知情同意。

发表同意书

不适用。

利益竞争

作者宣称他们没有竞争利益。

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参考文献

- Xu S, Liu X. Sequential treatment of chronic obstructive pulmonary disease concurrent with respiratory failure by high-flow nasal cannula therapy. *Am J Transl Res*. 2021;13(4):2831–9.
- Zheng M, Dong L, Hao Z, Wang S. Efficacy and safety of high-flow oxygen therapy application for chronic obstructive pulmonary disease with acute hypercapnic respiratory failure: a protocol for systematic review and meta-analysis. *Med*. 2021;100(15): e25489.
- Plotnikow GA, Accoce M, Fredes S, Tiribelli N, Setten M, Dorado J, Guaymas M, Ilutovich S, Rodriguez PO, Cesio CE, et al. High-flow oxygen therapy application in chronic obstructive pulmonary disease patients with acute hypercapnic respiratory failure: a multicenter study. *Crit Care Explor*. 2021;3(2): e0337.
- Johnson SA, Rupp AB, Rupp KL, Reddy S. Clinical outcomes and costs associated with procalcitonin utilization in hospitalized patients with pneumonia, heart failure, viral respiratory infection, or chronic obstructive pulmonary disease. *Intern Emerg Med*. 2021;16(3):677–86.
- Lin SH, He YP, Lian JJ, Chu CK. Procalcitonin kinetics to guide sequential invasive-noninvasive mechanical ventilation weaning in patients with acute exacerbation of chronic obstructive pulmonary disease and respiratory failure: procalcitonin's adjunct role. *Libyan J Med*. 2021;16(1): 1961382.
- Oli R, Mishra JK, Srivastava GN, Mishra S. Association of low body mass index with respiratory failure in chronic obstructive pulmonary disease. *J Nepal Health Res Counc*. 2021;19(1):135–9.
- Ellingsen I, Fondenes O, Overland B, Holmedahl NH. The severity of sleep hypoventilation in stable chronic obstructive pulmonary disease. *Sleep Breath*. 2021;25(1):243–9.
- Chen Y, Pan L. Nursing research on patients with chronic obstructive pulmonary disease and respiratory failure based on big data. *J Healthc Eng*. 2021;2021:2541751.
- Sohal AS, Anand A, Kaur P, Kaur H, Attri JP. Prospective comparative evaluation of noninvasive and invasive mechanical ventilation in patients of chronic obstructive pulmonary disease with acute respiratory failure type II. *Anesth Essays Res*. 2021;15(1):8–13.
- Pun BT, Badenes R, Heras La Calle G, Orun OM, Chen W, Raman R, Simpson BK, Wilson-Linville S, Hinojal Olmedillo B, de Vallejo la Cueva A, et al. Prevalence and risk factors for delirium in critically ill patients with COVID-19 (COVID-D): a multicentre cohort study. *Lancet Respir Med*. 2021;9(3):239–50.
- Mart MF, Williams Roberson S, Salas B, Pandharipande PP, Ely EW. Prevention and management of delirium in the intensive care unit. *Semin Respir Crit Care Med*. 2021;42(1):112–26.
- Cortes-Beringola A, Vicent L, Martin-Asenjo R, Puerto E, Dominguez-Perez L, Maruri R, Moreno G, Vidan MT, Fernando A, Bueno H. Diagnosis, prevention, and management of delirium in the intensive cardiac care unit. *Am Heart J*. 2021;232:164–76.
- Jin T, Jin Y, Lee SM. medication use and risk of delirium in mechanically ventilated patients. *Clin Nurs Res*. 2021;30(4):474–81.
- Habeeb-Allah A, Alshraideh JA. Delirium post-cardiac surgery: Incidence and associated factors. *Nurs Crit Care*. 2021;26(3):150–5.
- Zhang S, Han Y, Xiao Q, Li H, Wu Y. Effectiveness of bundle interventions on ICU delirium: a meta-analysis. *Crit Care Med*. 2021;49(2): 335–46.
- Staveski SL, Pickler RH, Khoury PR, Ollberding NJ, Donnellan AL, Mauney JA, Lincoln PA, Baird JD, Gilliland FL, Merritt AD, et al. Prevalence of ICU delirium in postoperative pediatric cardiac surgery patients. *Pediatr Crit Care Med*. 2021;22(1):68–78.
- Shivji S, Stabler SN, Boyce K, Haljan GJ, McGloin R. Management of delirium in a medical and surgical intensive care unit. *J Clin Pharm Ther*. 2021;46(3):669–76.
- Jianwen Y, Keke L. Sample size calculation methods. *Stat Decis Mak*. 2013;22(1):4–6.
- Juanjuan X, Li L, Yuexia L. Analysis of the influencing factors of delirium in patients with COPD combined with respiratory failure and mechanical ventilation. *Qilu Nurs J*. 2021;27(9):4–7.
- Lei W, Weining F, Yaping L. Effects of early activity on delirium and respiratory mechanics in patients with acute exacerbation of chronic obstructive pulmonary disease on mechanical ventilation. *Guizhou Med*. 2019;38(11):10–5.
- Ely EW, Inouye SK, Bernard GR, Gordon S, Francis J, May L, Truman B, Speroff T, Gautam S, Margolin R, et al. Delirium in mechanically ventilated patients: validity and reliability of the confusion assessment method for the intensive care unit (CAM-ICU). *JAMA*. 2001;286(21):2703–10.
- Slovic B. Calculated decisions: critical-care pain observation tool (CPOT). *Emerg Med Pract*. 2019;21(11):CD3–4.
- García-Grimshaw M, Chiquete E, Jiménez-Ruiz A, Vidal-Mayo JJ, Grajeda-González SL, Vargas-Martínez MLA, Toapanta-Yanchapaxi LN, Valdés-Ferrer SI, Chávez-Martínez OA, Marché-Fernández OA. Delirium and associated factors in a cohort of hospitalized patients with coronavirus disease 2019. *J Acad Consult Liaison Psychiatr*. 2022;63(1):3–13. <https://doi.org/10.1016/j.jacp.2021.06.008>.
- Casault C, Soo A, Lee CH, Couillard P, Niven D, Stelfox T, Fiest K. Sedation strategy and ICU delirium: a multicentre, population-based propensity score-matched cohort study. *BMJ Open*. 2021;11(7): e045087.
- Wang S, Hong Y, Li S, Kuriyama A, Zhao Y, Hu J, Luo A, Sun R. Effect of dex-medetomidine on delirium during sedation in adult patients in intensive care units: a systematic review and meta-analysis. *J Clin Anesth*. 2021;69: 110157.
- Chen H, Mo L, Hu H, Ou Y, Luo J. Risk factors of postoperative delirium after cardiac surgery: a meta-analysis. *J Cardiothorac Surg*. 2021;16(1): 113.
- Bode L, Fuchs S, Gehrke S, Ernst J, Schubert M, Nunez DG, von Kanel R, Boettger S. Assessing delirium with nursing care instruments: evaluation of the cognitive and associated domains. *Palliat Support Care*. 2021; 19(1):11–6.
- Cui F, Zhao W, Mu DL, Zhao X, Li XY, Wang DX, Jia HQ, Dai F, Meng L. Association between cerebral desaturation and postoperative delirium in thoracotomy with one-lung ventilation: a prospective cohort study. *Anesth Analg*. 2021;133(1):176–86.
- Romero N, Dube KM, Lupi KE, DeGrado JR. Evaluation of delirium in critically ill patients prescribed melatonin or ramelteon. *Ann Pharmacother*. 2021;55(11):1347–54.
- Szylińska A, Rotter I, Listewnik M, Lechowicz K, Brykczyński M, Dzidek S, Źukowski M, Kotfis K. Postoperative delirium in patients with chronic obstructive pulmonary disease after coronary artery bypass grafting. *Medicina*. 2020;56(7):342. <https://doi.org/10.3390/medic56070342>.
- Wu J, Yin Y, Jin M, Li B. The risk factors for postoperative delirium in adult patients after hip fracture surgery: a systematic review and meta-analysis. *Int J Geriatr Psychiatr*. 2021;36(1):3–14.
- Kerber K, Zangmeister J, McNett M. Relationship between delirium and ventilatory outcomes in the medical intensive care unit. *Crit Care Nur*. 2020; 40(2):24–31.
- Rollo E, Callea A, Brunetti V, Vollono C, Marotta J, Imperatori C, Frisullo G, Broccolini A, Della Marca G. Delirium in acute stroke: a prospective, cross-sectional, cohort study. *Eur J Neurol*. 2021;28(5):1590–600.
- Siebert FE. Postoperative delirium in the elderly surgical patient. *Anesthesiol Clin*. 2009;27(3):451–64. <https://doi.org/10.1016/j.anclin.2009.07.009>.
- Wilkinson ST, Chang SB, Foley T, Lipsey JR, MacKinnon DF. Delirium presenting as mania in a patient with end-stage COPD. *Am J Psychiatr*. 2014;171(8):821–4.
- Bilge EU, Kaya M, Senel GO, Unver S. The incidence of delirium at the postoperative intensive care unit in adult patients. *Turk J Anaesthesiol Reanim*. 2015;43(4):232–9.
- Tilouche N, Hassen MF, Ali HBS, Jaoued O, Gharbi R, El Atrous SS. Delirium in the intensive care unit: incidence, risk factors, and impact on outcome. *Indian J Crit Care Med*. 2018;22(3):144–9.
- Cui XP, Jing ZZ, Song JF, Zhang P. A retrospective study on risk factors associated with postoperative delirium in elderly patients with spinal operation. *Zhongguo Gu Shang*. 2019;32(6):549–54.
- Rothberg MB, Herzig SJ, Pekow PS, Avrunin J, Lagu T, Lindenauer PK. Association between sedating medications and delirium in older inpatients. *J Am Geriatr Soc*. 2013;61(6):923–30.

40. Tian H, Chen M, Yu W, Ma Q, Lu P, Zhang J, Jin Y, Wang M. Risk factors associated with postoperative intensive care unit delirium in patients undergoing invasive mechanical ventilation following acute exacerbation of chronic obstructive pulmonary disease. *J Int Med Res.* 2020;48(8):300060520946516.
41. Tan H, Zhou L, Wu S, Dong Q, Yang L, Xu J, Zhao S, Wang X, Yang H. Sub-jective strain of care experienced by pulmonary and critical care medical nurses when caring for patients with delirium: a cross-sectional study. *BMC Health Serv Res.* 2021;21(1):808.
42. Kotfis K, Williams Roberson S, Wilson J, Pun B, Ely EW, Jezowska I, Jezierska M, Dabrowski W. COVID-19: what do we need to know about ICU delirium during the SARS-CoV-2 pandemic? *Anaesthesiol Intensive Ther.* 2020;52(2):132–8.
43. Khan SH, Wang S, Harrawood A, Martinez S, Heiderscheit A, Chlan L, Perkins AJ, Tu W, Boustani M, Khan B. Decreasing delirium through music (DDM) in critically ill, mechanically ventilated patients in the intensive care unit: study protocol for a pilot randomized controlled trial. *Trials.* 2017;18(1):574.
44. Inouye SK, Westendorp RG, Saczynski JS. Delirium in elderly people. *Lancet.* 2014;383(9920):911–22.
45. Hsieh TT, Inouye SK, Oh ES. Delirium in the elderly. *Psychiatr Clin North Am.* 2018;41(1):1–17.
46. Hsieh TT, Inouye SK, Oh ES. Delirium in the Elderly. *Clin Geriatr Med.* 2020;36(2):183–99.
47. Janssen TL, Alberts AR, Hooft L, Mattace-Raso F, Mosk CA, van der Laan L. Prevention of postoperative delirium in elderly patients planned for elective surgery: systematic review and meta-analysis. *Clin Interv Aging.* 2019;14:1095–117.
48. Fong TG, Tulebaev SR, Inouye SK. Delirium in elderly adults: diagnosis, prevention and treatment. *Nat Rev Neurol.* 2009;5(4):210–20.
49. Witlox J, Eurelings LS, de Jonghe JF, Kalisvaart KJ, Eikelenboom P, van Gool WA. Delirium in elderly patients and the risk of postdischarge mortality, institutionalization, and dementia: a meta-analysis. *JAMA.* 2010;304(4):443–51.
50. Setters B, Solberg LM. Delirium. *Prim Care.* 2017;44(3):541–59.
51. Juanjuan X, Li L, Yuexia L. Analysis of influencing factors of delirium in COPD patients with respiratory failure undergoing mechanical ventilation. *Qilu J Nurs.* 2020;27(9):4–6.
52. Qiong W. Analysis of influencing factors of delirium in mechanically ventilated patients. *Chin J Nurs.* 2010;53(5):543–8.
53. Devlin JW, Skrobik Y, Gelinas C, Needham DM, Slooter AJC, Pandharipande PP, Watson PL, Weinhouse GL, Nunnally ME, Rochweg B, et al. Clinical practice guidelines for the prevention and management of pain, agitation/sedation, delirium, immobility, and sleep disruption in adult patients in the ICU. *Crit Care Med.* 2018;46(9):e825–73.
54. Harris J, Ramelet AS, van Dijk M, Pokorna P, Wielenga J, Tume L, Tibboel D, Ista E. Clinical recommendations for pain, sedation, withdrawal and delirium assessment in critically ill infants and children: an ESPNIC position statement for healthcare professionals. *Intensive Care Med.* 2016;42(6):972–86.
55. Hermes C, Acevedo-Nuevo M, Berry A, Kjellgren T, Negro A, Massarotto P. Gaps in pain, agitation and delirium management in intensive care: outputs from a nurse workshop. *Intensive Crit Care Nurs.* 2018;48:52–60.
56. Feast AR, White N, Lord K, Kupeli N, Vickerstaff V, Sampson EL. Pain and delirium in people with dementia in the acute general hospital setting. *Age Ageing.* 2018;47(6):841–6.
57. Jin Z, Hu J, Ma D. Postoperative delirium: perioperative assessment, risk reduction, and management. *Br J Anaesth.* 2020;125(4):492–504.
58. Devlin JW, Skrobik Y, Gelinas C, Needham DM, Slooter AJC, Pandharipande PP, Watson PL, Weinhouse GL, Nunnally ME, Rochweg B, et al. Executive summary: clinical practice guidelines for the prevention and management of pain, agitation/sedation, delirium, immobility, and sleep disruption in adult patients in the ICU. *Crit Care Med.* 2018;46(9):1532–48.
59. Wong IMJ, Thangavelautham S, Loh SCH, Ng SY, Murfin B, Shehabi Y. Sedation and Delirium in the Intensive Care Unit-A Practice-Based Approach. *Ann Acad Med Singap.* 2020;49(4):215–25.
60. Morandi A, Brummel NE, Ely EW. Sedation, delirium and mechanical ventilation: the “ABCDE” approach. *Curr Opin Crit Care.* 2011;17(1):43–9.
61. Liu X, Xie G, Zhang K, Song S, Song F, Jin Y, Fang X. Dexmedetomidine vs propofol sedation reduces delirium in patients after cardiac surgery: a meta-analysis with trial sequential analysis of randomized controlled trials. *J Crit Care.* 2017;38:190–6.
62. Sieber FE, Neufeld KJ, Gottschalk A, Bigelow GE, Oh ES, Rosenberg PB, Mears SC, Stewart KJ, Ouannes JP, Jaber M, et al. Effect of depth of sedation in older patients undergoing hip fracture repair on post-operative delirium: the stride randomized clinical trial. *JAMA Surg.* 2018;153(11):987–95.
63. Pandharipande PP, Pun BT, Herr DL, Maze M, Girard TD, Miller RR, Shintani AK, Thompson JL, Jackson JC, Deppen SA, et al. Effect of sedation with dexmedetomidine vs lorazepam on acute brain dysfunction in mechanically ventilated patients: the MENDS randomized controlled trial. *JAMA.* 2007;298(22):2644–53.
64. Jakob SM, Ruokonen E, Grounds RM, Sarapohja T, Garratt C, Pocock SJ, Bratty JR, Takala J. Dexmedetomidine for Long-Term Sedation I: dexmedetomidine vs midazolam or propofol for sedation during prolonged mechanical ventilation: two randomized controlled trials. *JAMA.* 2012;307(11):1151–60.
65. Riker RR, Shehabi Y, Bokesch PM, Ceraso D, Wisemandle W, Koura F, Whitten P, Margolis BD, Byrne DW, Ely EW, et al. Dexmedetomidine vs midazolam for sedation of critically ill patients: a randomized trial. *JAMA.* 2009;301(5):489–99.
66. Girard TD. Sedation, delirium, and cognitive function after critical illness. *Crit Care Clin.* 2018;34(4):585–98.
67. Pereira JV, Sanjanwala RM, Mohammed MK, Le ML, Arora RC. Dexmedetomidine versus propofol sedation in reducing delirium among older adults in the ICU: a systematic review and meta-analysis. *Eur J Anaesthesiol.* 2020;37(2):121–31.
68. Baron R, Binder A, Biniek R, Braune S, Buerkle H, Dall P, Demirakca S, Eckardt R, Eggers V, Eichler I, Fietze I. Evidence and consensus based guideline for the management of delirium, analgesia, and sedation in intensive care medicine. Revision 2015 (DAS-Guideline 2015)–short version. *GMS German Medical Science.* 2015;13.
69. Frontera JA. Delirium and sedation in the ICU. *Neurocrit Care.* 2011;14(3):463–74.
70. Deffland M, Spies C, Weiss B, Keller N, Jenny M, Kruppa J, Balzer F. Effects of pain, sedation and delirium monitoring on clinical and economic outcome: a retrospective study. *PLoS ONE.* 2020;15(9):e0234801.

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